

Department of CS&IT
PROGRAMME: BCA (Bachelor of Computer Applications)-
ISMA & MACT
SEMESTER – VI

Teaching Learning and Evaluation Plan

Course Information:

Course Code: 23BCA6M02

Course Title: IOS Application Development

Credits Units: 03

Contact Hours: 45

L-T-P-E: 3-0-0-3

IA: UE Weightage - 50: 50

Pass Marks (IA & UE) – NA&18

Aggregate Pass Marks: 40

UE Question Paper Marks: 50

Special Examination Fees: NA

Pre-requisite (if any):

1. Students should know the programming language such as Swift.
2. Students should know the basic concepts of MAC & IOS.

Course Facilitator (s): Prof. Diwakara Vasuman M S, Assistant Professor, School of CS & IT

Programme Outcomes (POs) and Programme Specific Outcomes (PSOs)

At the end of the programme, students will be able to	
PO 1	Computational Knowledge: Understand and apply mathematical foundation, computing and domain knowledge for the conceptualization of computing models from defined problems.
PO 2	Problem Analysis: Ability to identify, critically analyze and formulate complex computing problems using fundamentals of computer science and application domains.
PO 3	Design / Development of Solutions: Ability to transform complex business scenarios and contemporary issues into problems, investigate, understand and propose integrated solutions using emerging technologies.
PO 4	Conduct Investigations of Complex Computing Problems: Ability to devise and conduct experiments, interpret data and provide well-informed conclusions.
PO 5	Modern Tool Usage: Ability to select modern computing tools, skills and techniques necessary for innovative software solutions.
PO 6	Professional Ethics: Ability to apply and commit professional ethics and cyber regulations in a global economic environment.
PO 7	Life-long Learning: Recognize the need for and develop the ability to engage in continuous learning as a Computing professional.
PO 8	Project Management: Ability to understand management and computing principles with computing knowledge to manage projects in multidisciplinary environments.
PO 9	Communication Efficacy: Communicate effectively with the computing community as well

	as society by being able to comprehend effective documentation and presentations.
PO 10	Societal & Environmental Concern: Ability to recognize economical, environmental social, health, legal, ethical issues involved in the use of computer technology and other consequential responsibilities relevant to professional practice.
PO11	Individual & Team Work: Ability to work as a member or leader in diverse teams in a multidisciplinary environment.
PO12	Innovation and Entrepreneurship: Identify opportunities, entrepreneurship vision and use of innovative ideas to create value and wealth for the betterment of the individual and society.

PSO's

PSO1 Demonstrate a solid understanding of the basic concepts of IOS Application Development.

PSO2 Recognize various techniques and controllers for application development.

PSO3 Use the latest tools and techniques of interface builder in IOS.

PSO4 Design and develop solutions for real-world problems using Mobile and Cloud Computing Technologies

Course Objectives:

COB1.	To facilitate students to understand ios application development.
COB2	To help students to gain a basic understanding of IOS application development
COB3	To inculcate working knowledge of Android Studio development tool

Course Outcomes:

At the end of the course, students will be able to

Sl. No.	Course Outcome	Description	POs	Bloom's Taxonomy Level	Assessment Tool
1.	CO 1	Describe iPhone architecture and interface builder	PO01, PO02, PO03 PO04 PO05	L2	MOOC COURSE/ Presentation/ IT1/Mini Project/ESE
2.	CO 2	Describe Swift Basics.	PO01, PO02, PO03 PO04 PO05	L4	IT1/ MOOC COURSE/ Preparatory /Mini Project/ ESE
3.	CO 3	Analyze Layouts and controllers for iOS application Development.	PO01, PO02, PO03 PO04 PO05	L5	IT2/ MOOC COURSE/ Mini Project/ ESE
4.	CO 4	Distinguish UI Controllers	PO01, PO02, PO03 PO04 PO05	L2	IT2/ MOOC COURSE/ Mini Project/ ESE
5.	CO 5	Evaluate XML parsing & JSON parsing	PO01, PO02, PO03 PO04 PO05 ,PO8,PO11	L6	IT2/ MOOC COURSE /Report Writing on Mini project topic/application/ ESE

CO-PO/PSO Mapping:

Course Outcome	Bloom's Taxonomy Level	Program Outcomes (PO)												Program Specific Outcomes (PSO)			
		PO 01	PO 02	PO 03	PO 04	PO 05	PO 06	PO 07	PO 08	PO 09	PO 10	PO 11	PO 12	PS O 1	PS O 2	PSO 3 L4	PSO 4 L6
CO1	L2	3	3	3	3	3								2		1	2
CO2	L4	3	3	2	3	2								2	3	3	2
CO3	L5	3	3	3	3	3								2	3	3	2
CO4	L2	3	3	2	2	2								3	2	2	2
CO5	L6	3	2	3	3	2			3			3		3	3	3	3
CO Avg.		3	2.8	2.6	2.8	2.4			3			3		2.4	2.2	2.4	2.2

Course Contents:

Module	Details	Contact Hours
I	Module 1: Fundamentals Fundamentals: Overview of MAC OS and X-CODE, Introduction to iPhone Architecture, Essential COCOA Touch Classes, Interface Builder, Nib File, COCOA and MVC Framework, Overview of features of latest iOS	9
II	Module 2: Swift Basics Basics of objective c, Need of transformation from objective c to swift, Data types, variables, constants, operators, Decision making statements, looping, arrays, dictionaries, functions, enumerations, structure, classes, inheritance.	9
III	Module 3: iPhone Application Development Auto Layout, Views, Outlets and Actions, Different View Controller: single view Controller, Master Detail View Controller, Navigation View Controller, Managing Application Memory, Application delegate, Handling Keyboard Input	9
IV	Module4: UI Controller Label, Button, Text Field, Slider, Switch, Progress View, Page Control, Table View, Collection View, Image View, Text View, Web View, Map View, Date Picker, Picker View, Search Bar, Gestures, push notification, Image Picker, QR Code Scanner, Audio and Video, Accelerometer, Location service, 3D	9

	touch, attribute tracking, Making the app live, overview of watches.	
V	Module 5: Database Management SQLite, Web Services, JSON parsing, XML Parsing, Alamo fire	9

TEXT BOOKS: (Write in APA Format)

Text Books:	
TB1	iOS 10 Programming Fundamentals with Swift by Matt Neuburg - O'Reilly Media Pub
References Books:	
1	Building iPhone and iPad Electronic Projects - MikeWesterfield - O'Reilly Media Pub.
2	Head First iPhone and iPad Development, 2nd Edition - Dan Pilone, Tracey Pilone - O'Reilly Media
3	Beginning iPhone and iPad Web Apps - ChrisApers, Daniel Paterson - Apress Pub

Session-Wise Plan:

Pedagogy/Activity planned		Mode of Delivery	
P1	PPT presentation & Classroom interaction	M1	Synchronous – PPT
P2	Blended Learning with Hands-on	M2	Asynchronous/synchronous
A1	Activity-1:MOOC	M3	Synchronous - Hands-on
A2	Activity-2 Use case Studies	M4	Synchronous - Discussion

Relevant Website (RW):

RW-1	Mac OS https://developer.apple.com
RW-2	Swift tutorial, Swift Tutorial
RW-3	IOS Development using Swift: iOS Development Using Swift iOS Tutorial - Javatpoint
RW-4	IOS UI Controller iOS UIViewController - Javatpoint
RW-5	Core data & database iOS Managed Object Model - Javatpoint

Video Links (VL):

VL-1	Introduction to MAC OS
VL-2	How to use Xcode in Swift https://youtu.be/N-ntKJdVNBs?si=yPUd4KCryjFQLM9a
VL-3	OOPS Concept in Swift with Snippets

	https://youtu.be/XdZUVmqlkJE?si=zAsz9hiFBAmHeN4F
VL-4	Getting started with iOS https://youtu.be/UC53g9VVDiI?si=OBIQYbDffGrhIIgy
VL-5	UI Controllers https://youtu.be/5wRsNSsVMZM?si=SWU-baNIGu0lgBwF
VL-6	Navigate the view controller https://youtu.be/ib4E2L-jjQU?si=Vyidbn-GHowlEt0v
VL-7	SQL Lite with iOS https://youtu.be/MQmEf-_v_2g?si=rOmuZztDGgBexysF

MOOC Courses (MC):

Sr.No.	Platform	Topic	CO	Link	Duration
MC-1	Coursera	Create a new iOS project and build your first iOS app	CO1	https://www.coursera.org/specializations/swift-5-ios-app-developer	1hr 6m
MC-2	Coursera	Use programming commands to allow for user interaction	CO2	https://www.coursera.org/specializations/swift-5-ios-app-developer	1hr 49m
MC-3	Coursera	How to create memory-efficient tables and display data in a list	CO3	https://www.coursera.org/specializations/swift-5-ios-app-developer	3hr 55m
MC-4	Coursera	How to model data and create database relationships	CO4	https://www.coursera.org/specializations/swift-5-ios-app-developer	7 hour 44 min
MC-5	Coursera	Write and execute unit tests to keep your code error free	CO5	https://www.coursera.org/specializations/swift-5-ios-app-developer	2 hours
Total MOOC Course integration with Certification					15.5 hours

Hardware/Software/Equipment (HSE):

HSE-1	Xcode - https://developer.apple.com/xcode/
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Session-wise Planning:

Module	Session	Topic	Study Materials	TB/ RB/ OR/ RW/ VL/	Pedagogy/ Activity planned	CO	Mode delivery of	Remarks
Bride Course	Zero Lecture	Introduction to MAC OS & XCode	https://drive.google.com/drive/folders/1yDIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link	TB-1 RB-1 VL-1,2	Blended Learning Approach	CO1	Synchronous-PPT Asynchronous-Video	Yes
I	1	Fundamentals of MAC OS	https://drive.google.com/drive/folders/1yDIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link		PPT presentation & Classroom interaction		Synchronous-PPT Asynchronous-Video	Yes
	2	Overview of X-CODE,	https://drive.google.com/drive/folders/1yDIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link		PPT presentation & Classroom interaction		Interactive Lecture Synchronous-Video	Yes
	3	Introduction to iPhone Architecture,	https://drive.google.com/drive/folders/1yDIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link		PPT presentation & Classroom interaction		Synchronous-PPT Asynchronous-Video	Yes
	4	Essential COCOA Touch Classes,	https://drive.google.com/drive/folders/1yDIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link		PPT presentation & Classroom interaction		Synchronous-PPT Asynchronous-Video	Yes

			DP_dYtqlq2 NHDMnr?u sp=drive_lin k					
	5	Interface Builder,	https://drive .google.co m/drive/fold ers/1yDIIKci U4PUjNC6z DP_dYtqlq2 NHDMnr?u sp=drive_lin k		PPT presentation & Classroom interaction		Synchronous-PPT Asynchronous- Video	Yes
	6	Nib File,	https://drive .google.co m/drive/fold ers/1yDIIKci U4PUjNC6z DP_dYtqlq2 NHDMnr?u sp=drive_lin k		PPT presentation & Classroom interaction		Synchronous-PPT Asynchronous- Video	Yes
	7-8	COCOA and MVC Framework,	https://drive .google.co m/drive/fold ers/1yDIIKci U4PUjNC6z DP_dYtqlq2 NHDMnr?u sp=drive_lin k		PPT presentation & Classroom interaction		Synchronous-PPT Asynchronous- Video	Yes
	9	Overview of features of latest iOS	https://drive .google.co m/drive/fold ers/1yDIIKci U4PUjNC6z DP_dYtqlq2 NHDMnr?u sp=drive_lin k		Interactive & Assessment with Report		Synchronous- Discussion, Suggestions Assessment Certification on Completion	Yes
II	10	Basics of objective c,	https://drive .google.co m/drive/fold ers/1yDIIKci U4PUjNC6z DP_dYtqlq2 NHDMnr?u sp=drive_lin k	TB-1 RB- 2 VL- 1,2	PPT presentation & Classroom interaction	CO2	Synchronous- Interactive Video Lecture with Quiz	Yes
	11	Need of transformat ion from objective c to swift,	https://drive .google.co m/drive/fold ers/1yDIIKci U4PUjNC6z DP_dYtqlq2 NHDMnr?u		PPT presentation & Classroom interaction		Synchronous- Interactive Video Lecture	Yes

		sp=drive_link					
12	Data types, variables,	https://drive.google.com/drive/folders/1yDIIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link		PPT presentation & Classroom interaction		Synchronous - PPT	Practical demonstration done continued in next class
13	constants, operators,	https://drive.google.com/drive/folders/1yDIIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link		PPT presentation & Classroom interaction		Synchronous - PPT	Yes
14	Decision making statements, looping,	https://drive.google.com/drive/folders/1yDIIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link		PPT presentation & Classroom interaction		Synchronous - PPT	Yes
15	arrays,	https://drive.google.com/drive/folders/1yDIIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link		PPT presentation & Classroom interaction		Synchronous-Interactive Video Lecture	Yes
16	dictionaries , functions,	https://drive.google.com/drive/folders/1yDIIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link		Open & Close Discussion, Query Resolution, Monitoring		Synchronous-PPT Asynchronous-Video	Yes
17	enumerations, structure,	https://drive.google.com/drive/folders/1yDIIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link		PPT presentation & Classroom interaction		Synchronous-PPT Asynchronous-Video	Yes
		https://drive					Yes

	18	classes, inheritance	.google.com/drive/folders/1yDIIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link		PPT Presentation & Classroom interaction		Synchronous-Discussion, Suggestions Assessment Certification on Completion	
III	19	Auto Layout,	https://drive.google.com/drive/folders/1yDIIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link	TB-1 RB - 2,3 VL - 3	PPT Presentation & Demonstration	CO3	Synchronous-PPT Asynchronous-Video	Practical demonstration done continued in next class
	20	Views,	https://drive.google.com/drive/folders/1yDIIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link		PPT Presentation & Classroom interaction		Synchronous-PPT Asynchronous-Video	Yes
	21	Outlets and Actions,	https://drive.google.com/drive/folders/1yDIIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link		PPT Presentation & Demonstration		Synchronous-PPT Asynchronous-Video	Yes
	22	single view Controller,	https://drive.google.com/drive/folders/1yDIIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link		PPT Presentation & Classroom interaction		Synchronous-PPT Lecture	Yes
	23	Master-Detail View Controller,	https://drive.google.com/drive/folders/1yDIIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link		PPT Presentation & Demonstration		Synchronous-PPT Asynchronous-Video	Yes
	24	Navigation	https://drive.google.com/drive/folders/1yDIIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link		PPT		Synchronous-PPT	Yes

		View Controller,	m/drive/folders/1yDIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link		Presentation with Comparative Analysis of IDS & NIDS		Asynchronous-Video	
	25	Managing Application Memory,	https://drive.google.com/drive/folders/1yDIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link		PPT Presentation & Demonstration		Synchronous-PPT Asynchronous-Quiz	Yes
	26	Application delegate,	https://drive.google.com/drive/folders/1yDIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link		PPT presentation & Classroom interaction		Synchronous-PPT & Interactive Video	Yes
	27	Handling Keyboard Input	https://drive.google.com/drive/folders/1yDIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link		Interactive & Assessment with Report		Assessment Certification with Report	Yes
IV	28	Label, Button, Text Field, Slider,	https://drive.google.com/drive/folders/1yDIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link	TB – 1 RB- 1 VL – 3,4	PPT Presentation & Classroom interaction	CO4	Synchronous-PPT Lecture with interaction	Practical demonstration done continued in next class
	29	Switch, Progress View, Page Control,	https://drive.google.com/drive/folders/1yDIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link		PPT Presentation & Demonstration		Synchronous-PPT Lecture with demo	Yes
	30	Table View,	https://drive.google.com/drive/fold		PPT Presentation &		Synchronous-PPT Asynchronous-	Yes

		Collection View, Image View,	ers/1yDIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link		Classroom Interaction		Video	
31	Text View, Web View, Map View,	https://drive.google.com/drive/folders/1yDIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link			PPT Presentation & Comparative Analysis		Synchronous-PPT Asynchronous-Video	Yes
32	Date Picker, Picker View, Search Bar,	https://drive.google.com/drive/folders/1yDIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link			Classroom interaction with live demo		Synchronous-PPT & Interactive Video with Quiz	Practical demonstration done continued in next class
33	Gestures, push notification, Image Picker, .	https://drive.google.com/drive/folders/1yDIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link			PPT Presentation & Classroom Interaction		Synchronous-PPT Asynchronous-Video	Yes
34	QR Code Scanner, Audio and Video, Accelerometer,	https://drive.google.com/drive/folders/1yDIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link			Classroom discussion		Synchronous-Discussion, Suggestions Clarification	Yes
35	Location service, 3D touch, attribute tracking,	https://drive.google.com/drive/folders/1yDIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link			Interactive & Assessment with Report		Assessment Certification with Report	Yes
36	Making the app live, overview of	https://drive.google.com/drive/folders/1yDIKciU4PUjNC6z			Open & Close Discussion, Query Resolution,		PPT Assessment/Execution/ Report Submission	Yes

		watches	DP_dYtqlq2 NHDMnr?u sp=drive_lin k		Monitoring			
V	37	Overview of Database Managem ent:	https://drive .google.co m/drive/fold ers/1yDIKci U4PUjNC6z DP_dYtqlq2 NHDMnr?u sp=drive_lin k	TB-2 RB- 2,3 VL – 3,4	PPT & Explanation	CO5	Synchronous- PPT Presentation	Yes
	38	Introductio n Database Managem ent using Swift	https://drive .google.co m/drive/fold ers/1yDIKci U4PUjNC6z DP_dYtqlq2 NHDMnr?u sp=drive_lin k		PPT Presentation & Classroom interaction		Synchronous-PPT Asynchronous- Video	Yes
	39	SQLite,	https://drive .google.co m/drive/fold ers/1yDIKci U4PUjNC6z DP_dYtqlq2 NHDMnr?u sp=drive_lin k		Video Demonstration with Student Interaction		Synchronous-PPT & Interactive Video with Quiz	Yes
	40	Web Services,	https://drive .google.co m/drive/fold ers/1yDIKci U4PUjNC6z DP_dYtqlq2 NHDMnr?u sp=drive_lin k		Video Demonstration with Student Interaction		Synchronous-PPT & Interactive Video	Yes
	41	JSON parsing	https://drive .google.co m/drive/fold ers/1yDIKci U4PUjNC6z DP_dYtqlq2 NHDMnr?u sp=drive_lin k		PPT Presentation & Classroom interaction		Synchronous-PPT Asynchronous- Video	Yes
	42	XML Parsing,	https://drive .google.co m/drive/fold ers/1yDIKci U4PUjNC6z DP_dYtqlq2		PPT Presentation & Classroom interaction		Synchronous-PPT Asynchronous- Video	Yes

			NHDMnr?usp=drive_link					
43	Alamo fire		https://drive.google.com/drive/folders/1yDIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link		PPT Presentation & Classroom interaction		Synchronous live interaction with student and sharing and suggesting writing skills for reports.	Yes
44	Activity1: Introduction to Android Mobile Application Development		https://drive.google.com/drive/folders/1yDIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link		Interactive & Assessment with Report		Assessment Certification with Report	Yes
45	Activity2 Mini Project App review on Xcode platform		https://drive.google.com/drive/folders/1yDIKciU4PUjNC6zDP_dYtqlq2NHDMnr?usp=drive_link		Open & Close Discussion, Query Resolution, Monitoring		Asynchronous-Video, PPT Assessment Certification on Completion	Yes

Assessment Scheme: CA: ESE - 50:50

Assessment Scheme: IA: UE - 50:50

Sl. No.	Assessment Instrument	Formative/ Summative	Frequency	Weightage (%)	CO
1.	Activity 1-MOOC/Use Case Studies	Formative	1	15	CO1,CO2,CO3,CO4,CO5
2.	Activity 2-Mini Project		1	15	CO1,CO2,CO3,CO4,CO5
3.	Internal Test1		1	7.5	CO1, CO2,CO3
4.	Internal Test2	Formative	1	7.5	CO3,CO4,CO5
5.	Class participation		continuous	5	**5 MARKS based on class participation, activities submission on time and test attendance
6.	End Semester Exam	Summative	1	50	CO1, CO2, CO3, CO4,CO5
	Total			100	

Others Details:

1. Class Participation – Punctuality, Frequency, and Level of Interaction, Contribution to the Learning Process (1 – Poor/Below 75%, 2 – Good/75-80%, 3 – Very Good/80-90%, 4 and above – Excellent/Above 90%)
2. Review Paper – Parameters for Evaluation:
 - a. Stated objectives
 - b. Provides evidence
 - c. Critical analysis of evidence
 - d. Global self awareness
 - e. Perspective taking
 - f. Cultural diversity
3. Discipline – The students are expected to be punctual and regular in attending classes, submitting assignments and activities. They should be decently dressed all the time. In their interactions, they should be respectful and sensitive to everyone.
4. Chamber Consultation Hours – 11.00 am to 12.00 noon (Tentative)
5. Pre Reading – The readings/ references given in session-wise plan should be read by students before coming to class to have effective and meaningful discussion on the topic.
6. Participation in Co-curricular and Extra-Curricular Activities – Attendance for the course on Multidisciplinary Writing is compulsory as it will help in writing Review Essays

Bloom's Taxonomy-Revised

LEVEL	DESCRIPTION	MEANING	ACTION VERBS
6	Creating	Can the student create a new product or POV?	Assemble, construct, create, change, combine, compose, design, develop, formulate, invent, modify, organize, propose, theorize, write
5	Evaluating	Can the student justify a stand or decision?	Appraise, agree, assess, argue, conclude, decide, defend, judge, prioritize, prove, rate, recommend, select, support, value
4	Analyzing	Can the student distinguish between different parts?	Contrast, compare, criticize, differentiate, discriminate, dissect, distinguish, examine, experiment, operate, question, simplify, test
3	Applying	Can the student use information in a new way?	Choose, demonstrate, dramatize, employ, illustrate, interpret, schedule, sketch, solve, use
2	Understanding	Can the student explain ideas and concepts?	Classify, describe, discuss, explain, identify, infer, locate, outline, paraphrase, recognize, report, summarize, select, translate
1	Remembering	Can the student recall or remember information?	Define, duplicate, find, list, label, match, memorize, name, omit, recall, repeat, state, spell, tell

Signature of the Faculty

Signature of the HoD